Experiment: 10

Problem Statement:-

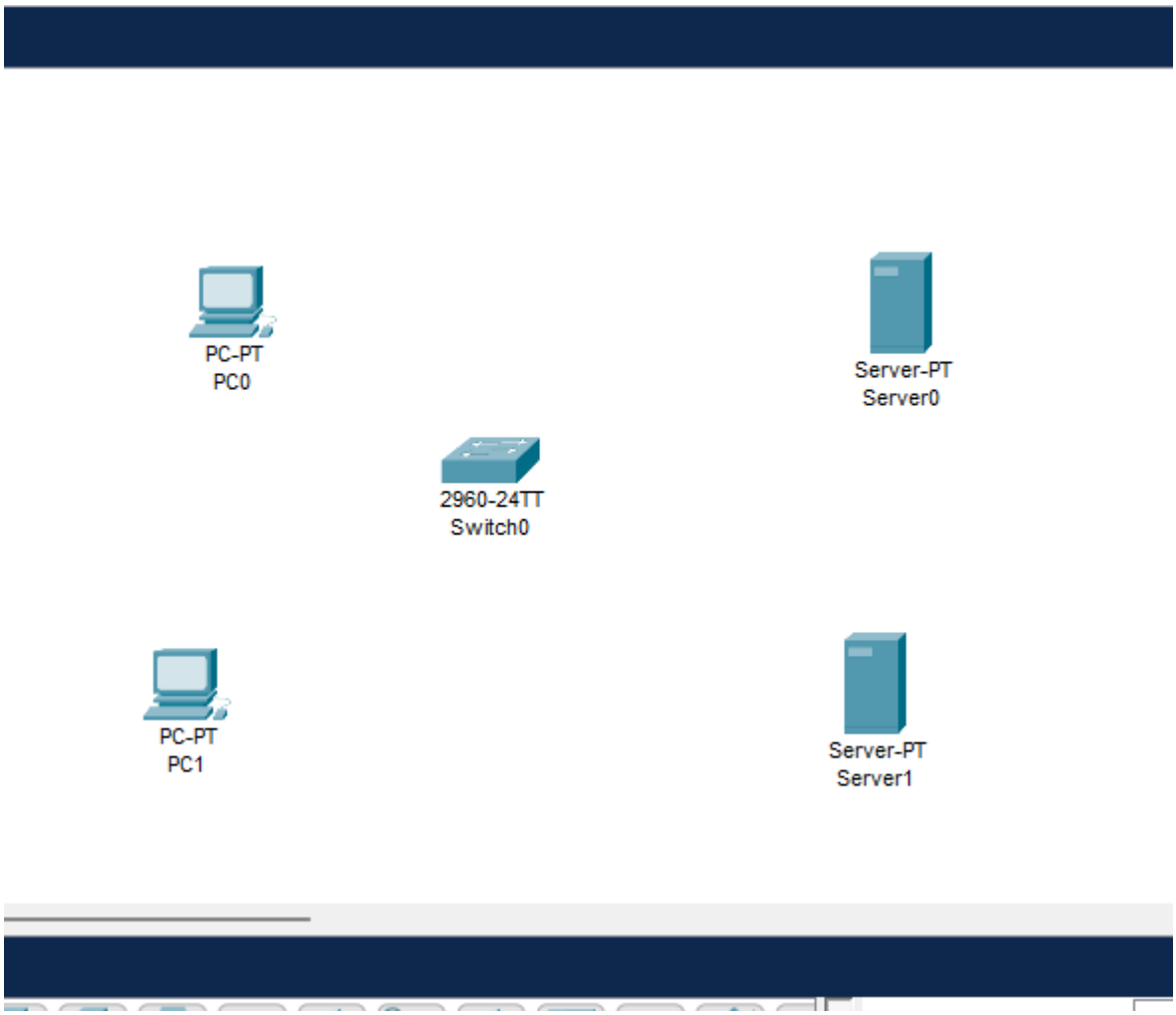
Implement a DNS (Domain Name System) and DDNS (Dynamic Domain Name System) simulation to demonstrate domain name resolution and dynamic updates. Create a simple client-server application within Cisco Packet Tracer that queries the DNS for domain names and updates DNS records dynamically.

Aim:- **DNS and DDNS Simulation: Domain Name Resolution and Dynamic Updates**

Procedure:-

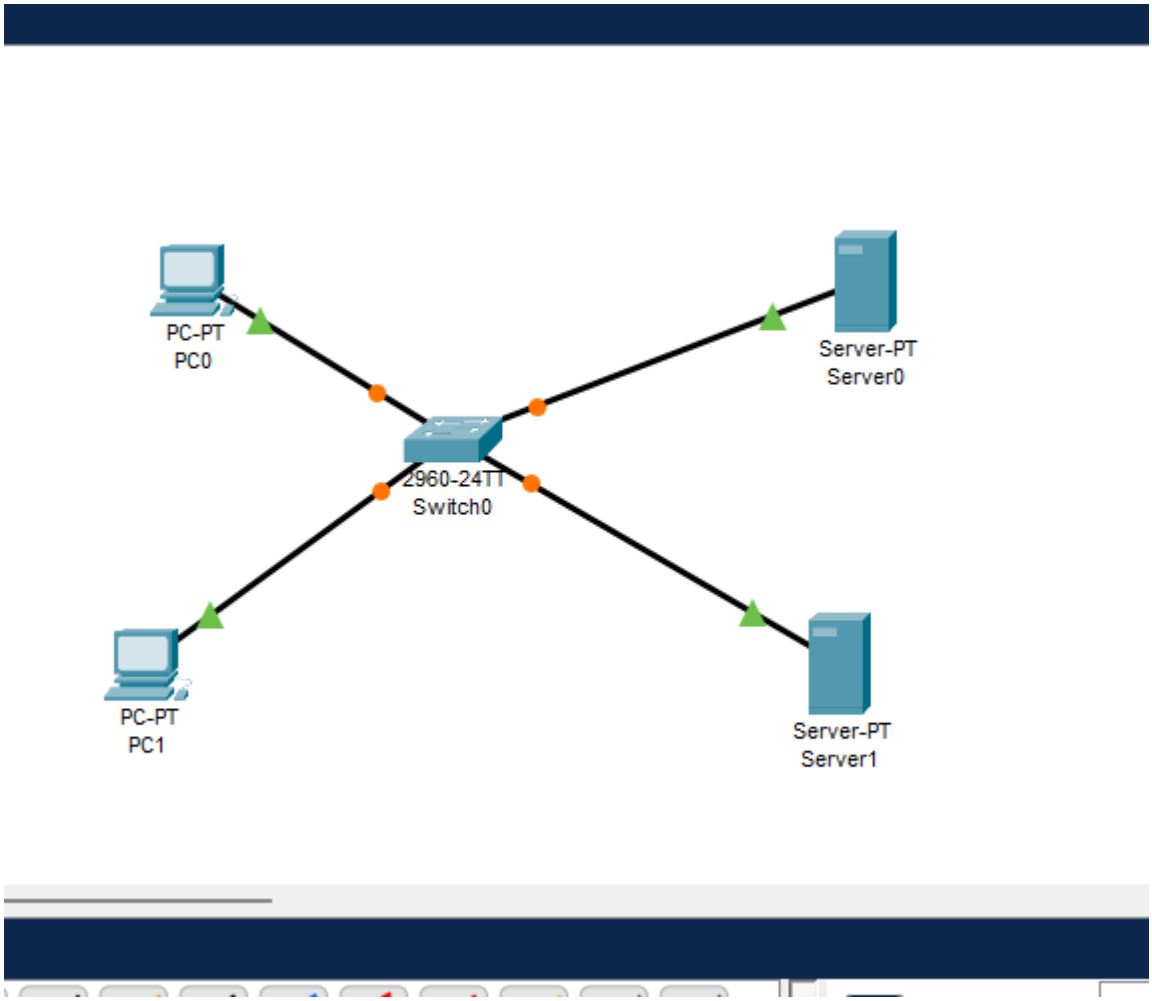
PART 1: Setup Network Topology

Step 1: Add Devices to Workspace



Step 2: Connect All Devices

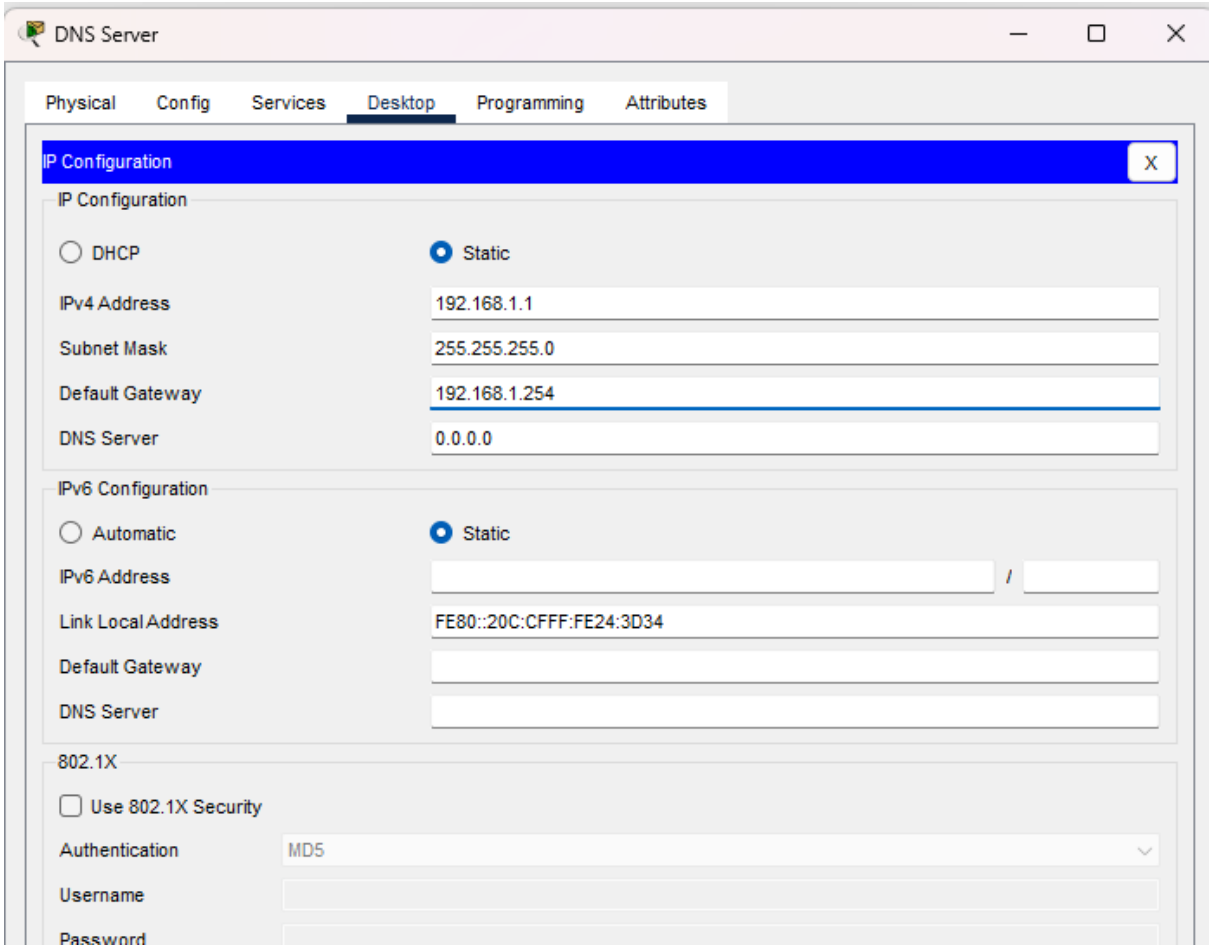
Use **Straight-Through Copper Cable:**



PART 2: Configure IP Addresses

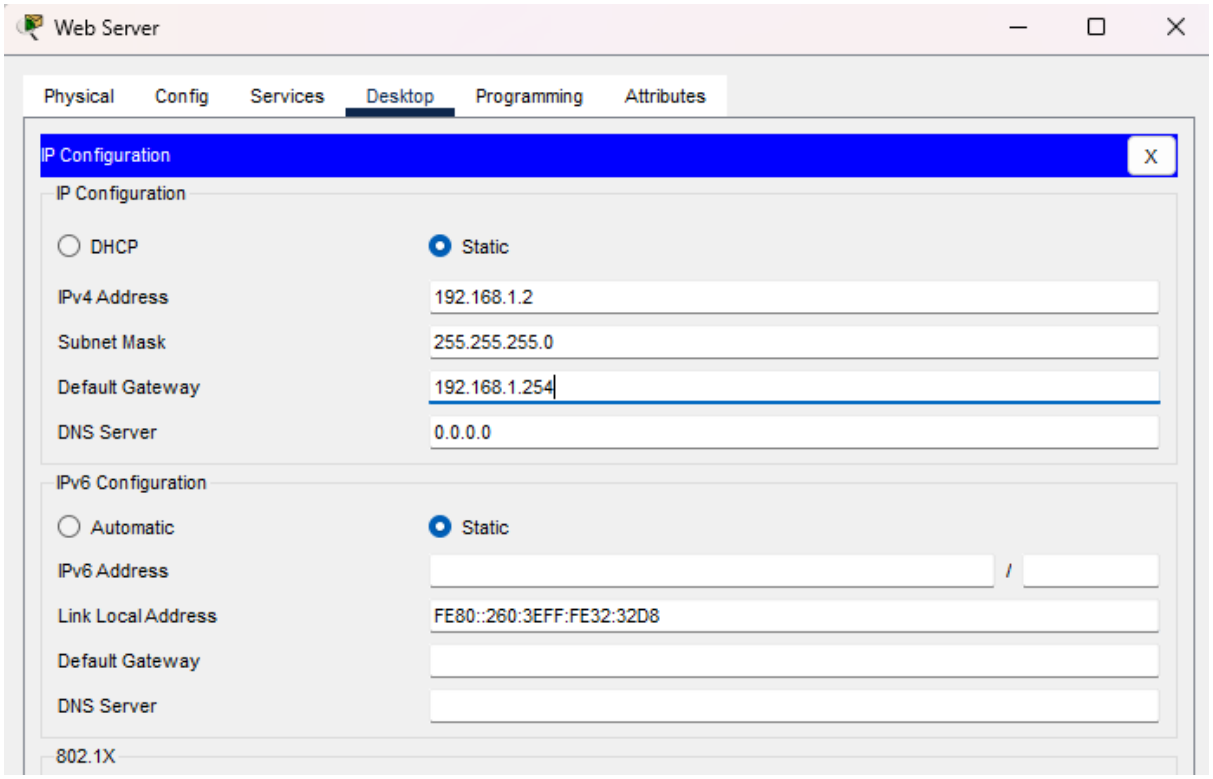
Step 3: Assign IP to DNS Server

1. Double-click **DNS Server**
2. Go to **Desktop → IP Configuration**
3. Enter:

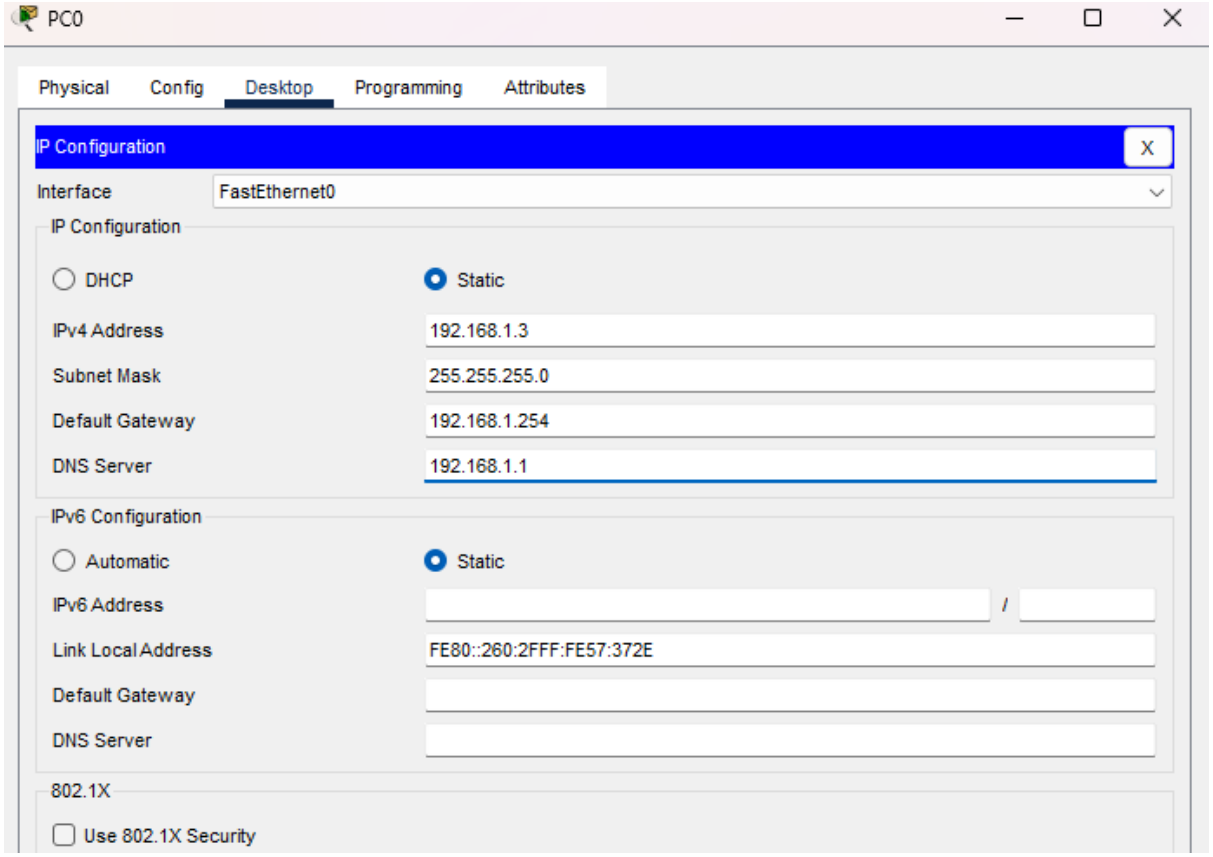


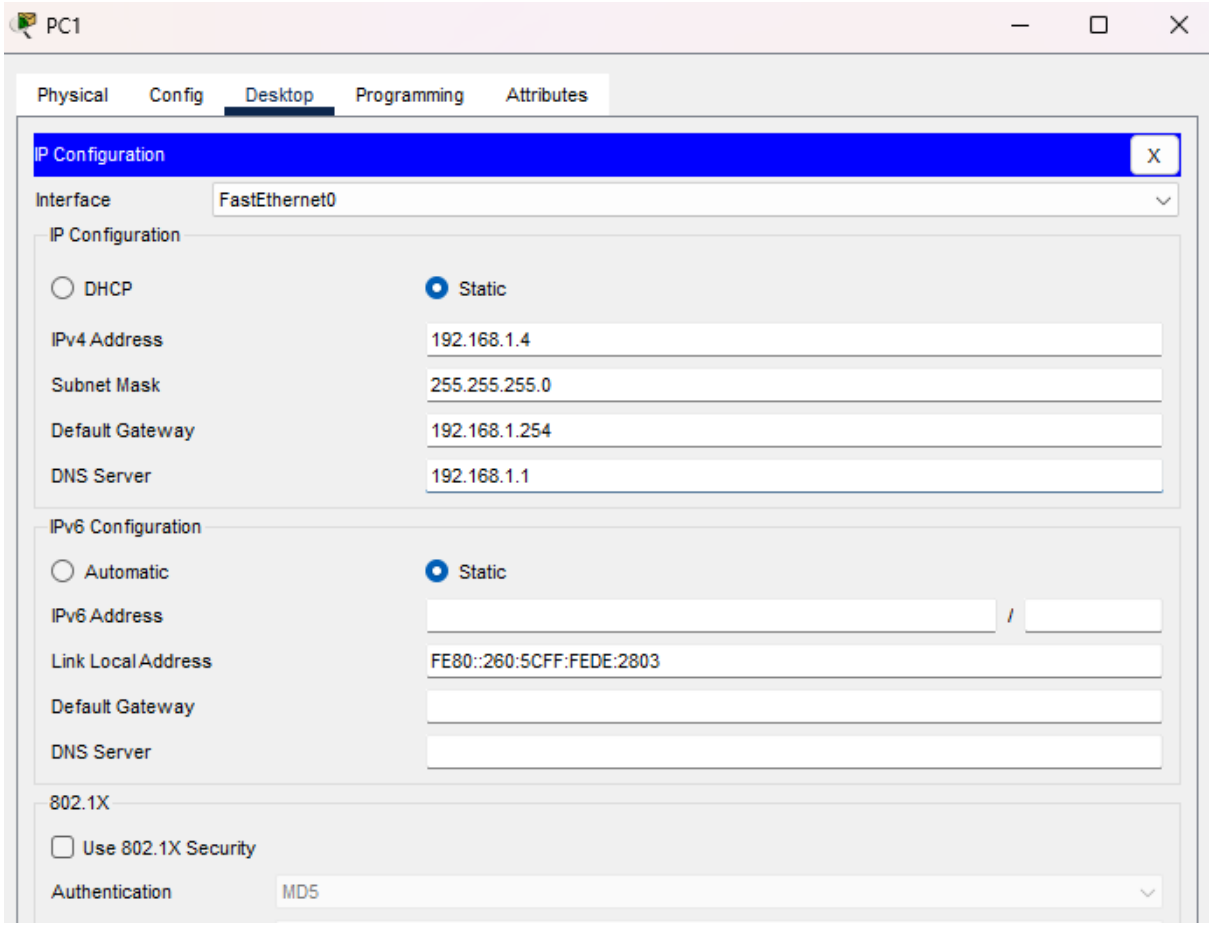
Step 4: Assign IP to Web Server

1. Double-click **Web Server**
2. Go to **Desktop → IP Configuration**
3. Enter:



Step 5: Assign IP to PCs

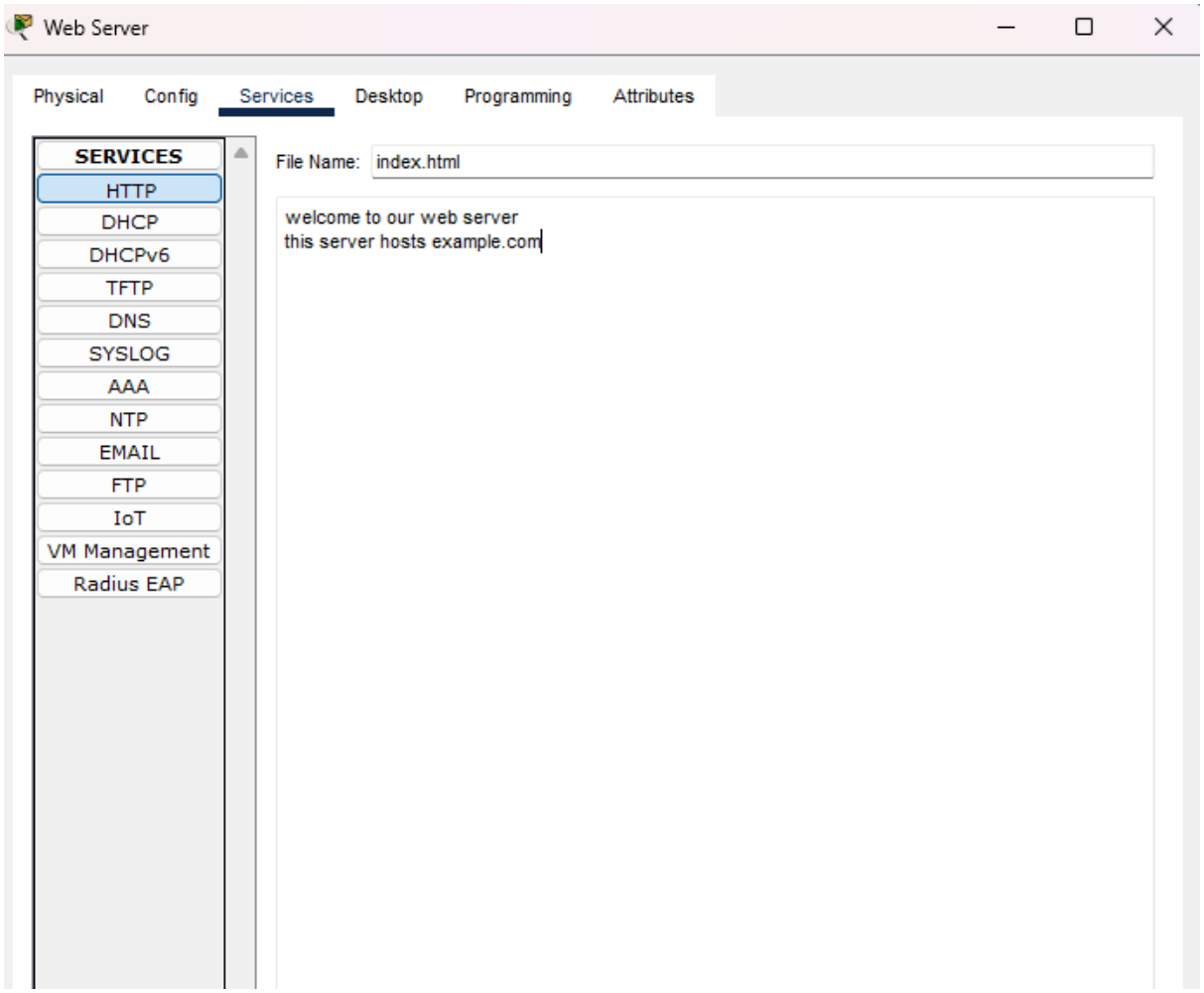




PART 3: Configure Web Server

Step 7: Enable HTTP on Web Server

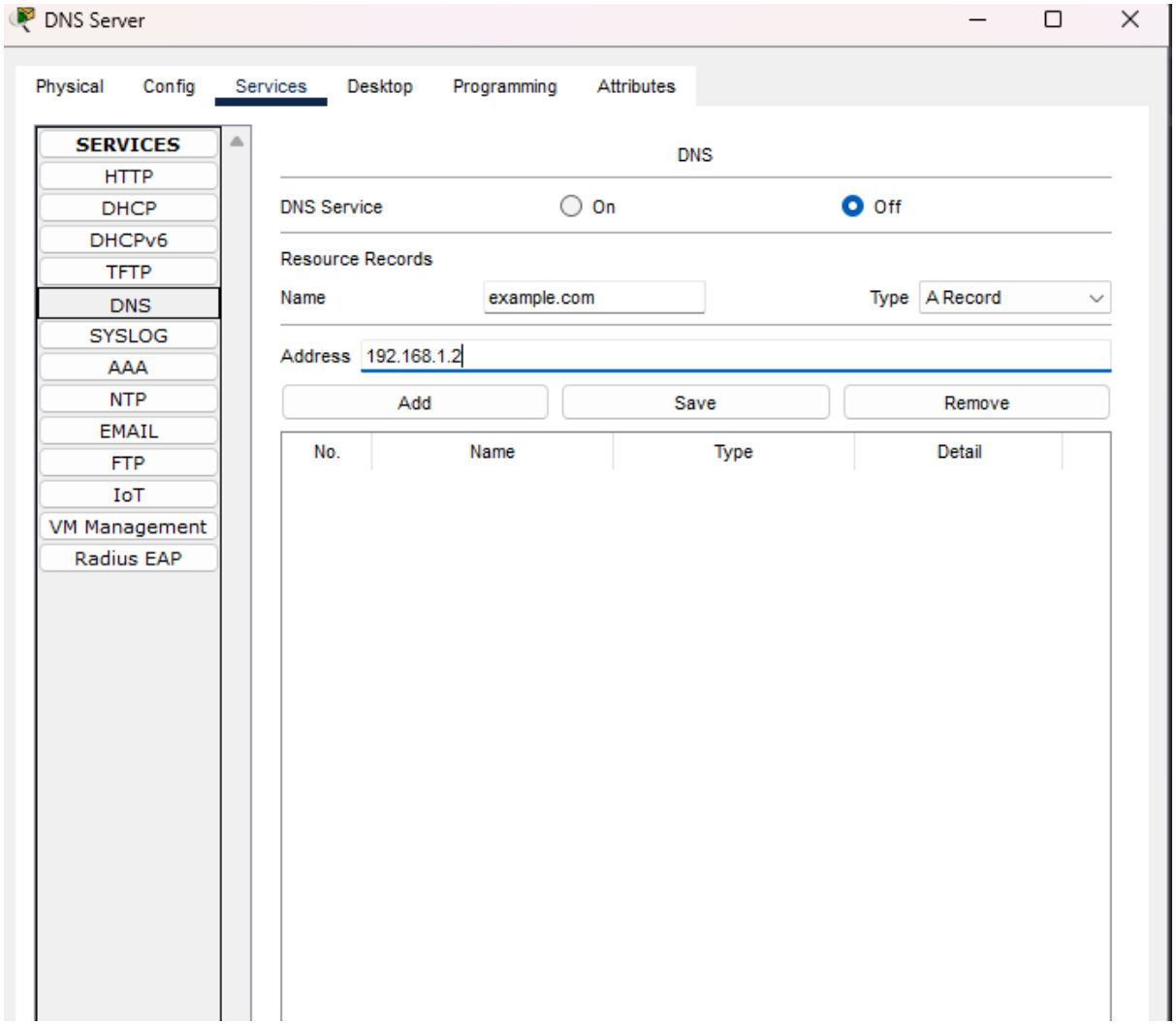
1. Double-click **Web Server**
2. Go to **Services** tab
3. Click **HTTP**
4. Turn Service **ON** ☑
5. Edit default webpage to say:

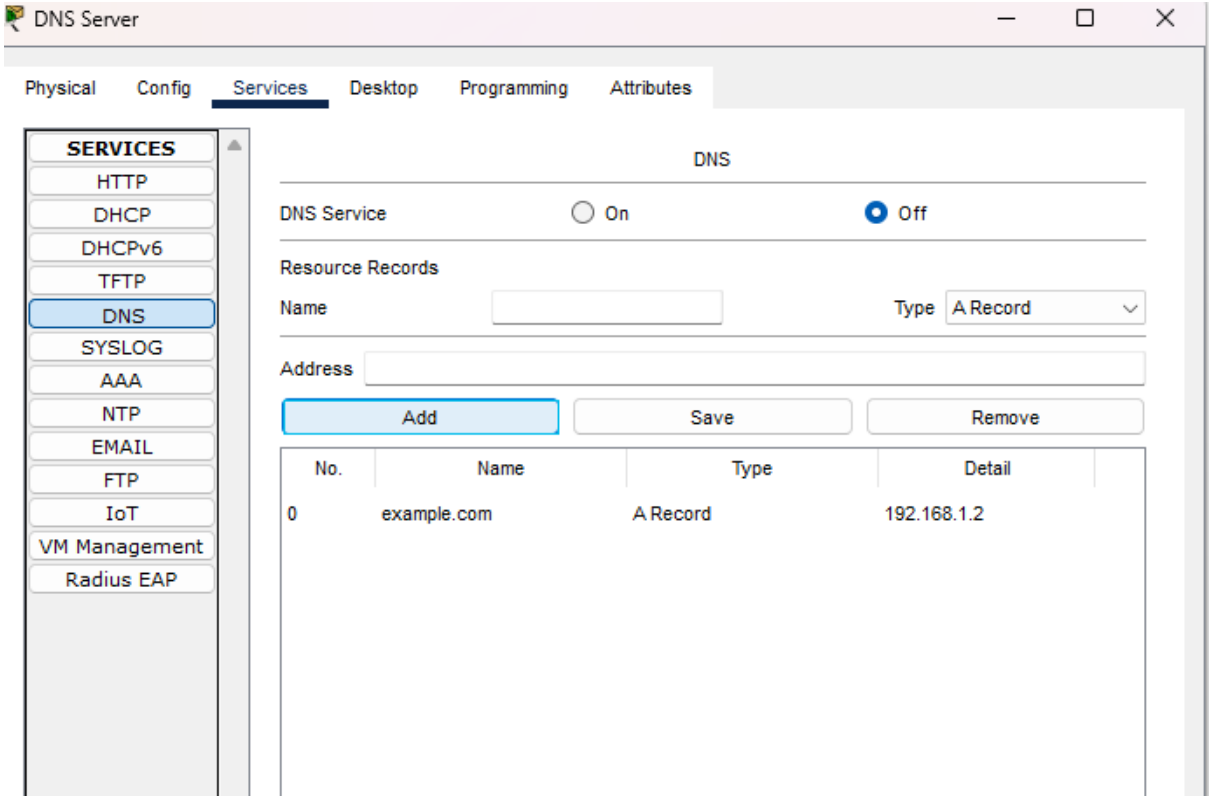


PART 4: Configure DNS Server

Step 8: Enable DNS Service

1. Double-click **DNS Server**
2. Go to **Services** tab
3. Click **DNS** on left sidebar
4. Toggle Service to **ON** ☑

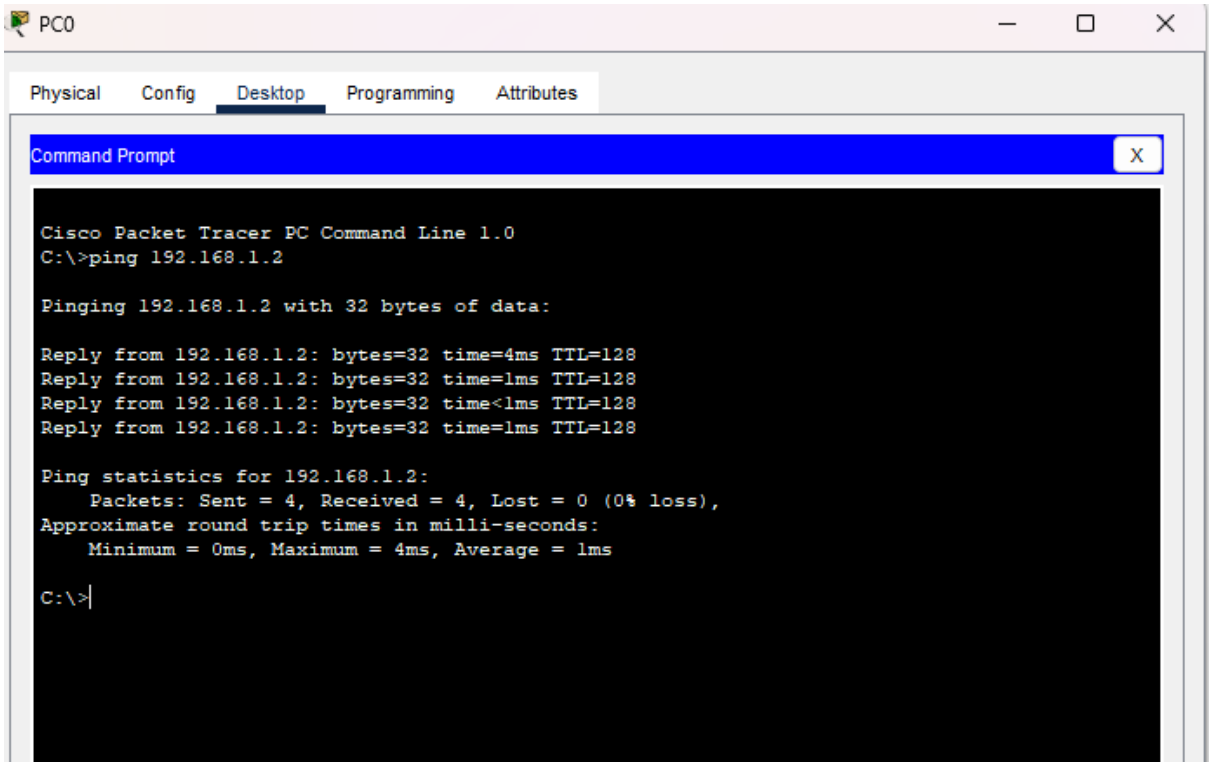

DNS Server
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
DNS
DNS Service
On
Off
Resource Records
Name
example.com
Type
A Record
Address
192.168.1.2
Add
Save
Remove
No.
Name
Type
Detail



PART 5: Test Network Connectivity

Step 10: Ping Test from PC0

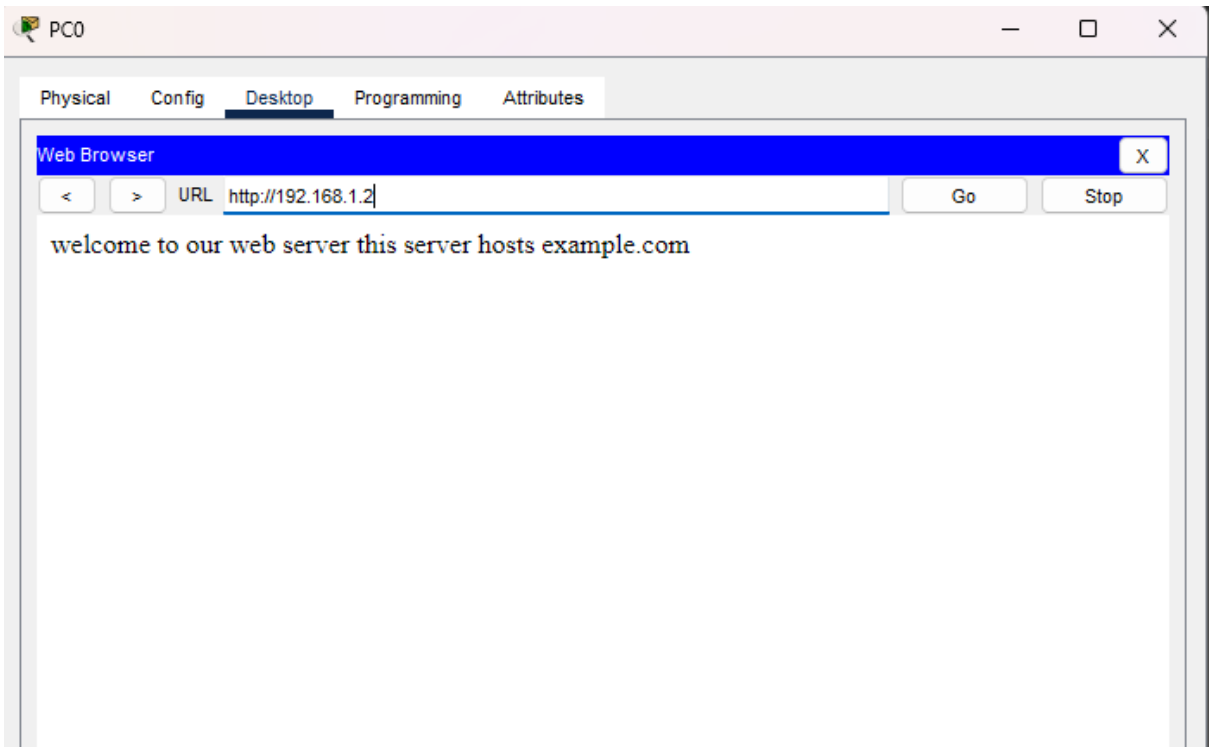
1. Double-click **PC0**
2. Go to **Desktop → Command Prompt**
3. Type:



PART 6: Query DNS from Clients

Step 11: Test DNS via Web Browser on PC0

1. Double-click **PC0**
2. Go to **Desktop → Web Browser**
3. Type in address bar:



Step 12: Test DNS via nslookup on PC1

1. Double-click **PC1**
2. Go to **Desktop → Command Prompt**
3. Type:

```
C:\>nslookup example.com

Server: [192.168.1.1]
Address:  192.168.1.1

Non-authoritative answer:
Name:   example.com
Address:   192.168.1.2

C:\>
```

PART 7: Update DNS Records

Step 13: Update example.com Record

1. Double-click **DNS Server**
2. Go to **Services → DNS**
3. Click on **example.com** in the list
4. Change address to: